Phase 3: Design Rationale and Self Audit

Design Rationale

I chose Case B because it provided a clear opportunity to reduce bias through interaction design, not just cosmetic changes. In the original Gapminder view, animation, logarithmic scaling, and large bubble salience strongly shape interpretation. A redesign could therefore reduce bias while preserving the same variables and overall topic.

My design followed the Visualization Design Process framework. In Stage 1, I approached the task as problem framing rather than a style change. This phase showed that the main issues were not aesthetic: autoplay encouraged a smooth progress narrative, the default x-axis hid an important log-scale choice, and fluid animation overstated continuity in the historical data. This led to a clear redesign goal: keep the same topic and variables, but replace cinematic persuasion with deliberate comparison.

In Stage 2, I translated the identified biases into concrete design choices.
The first decision was to replace autoplay with small multiples showing fixed comparison years. This follows lecture guidance on multiple-object tracking limits and the trade-off between animation and small multiples for temporal comparison.

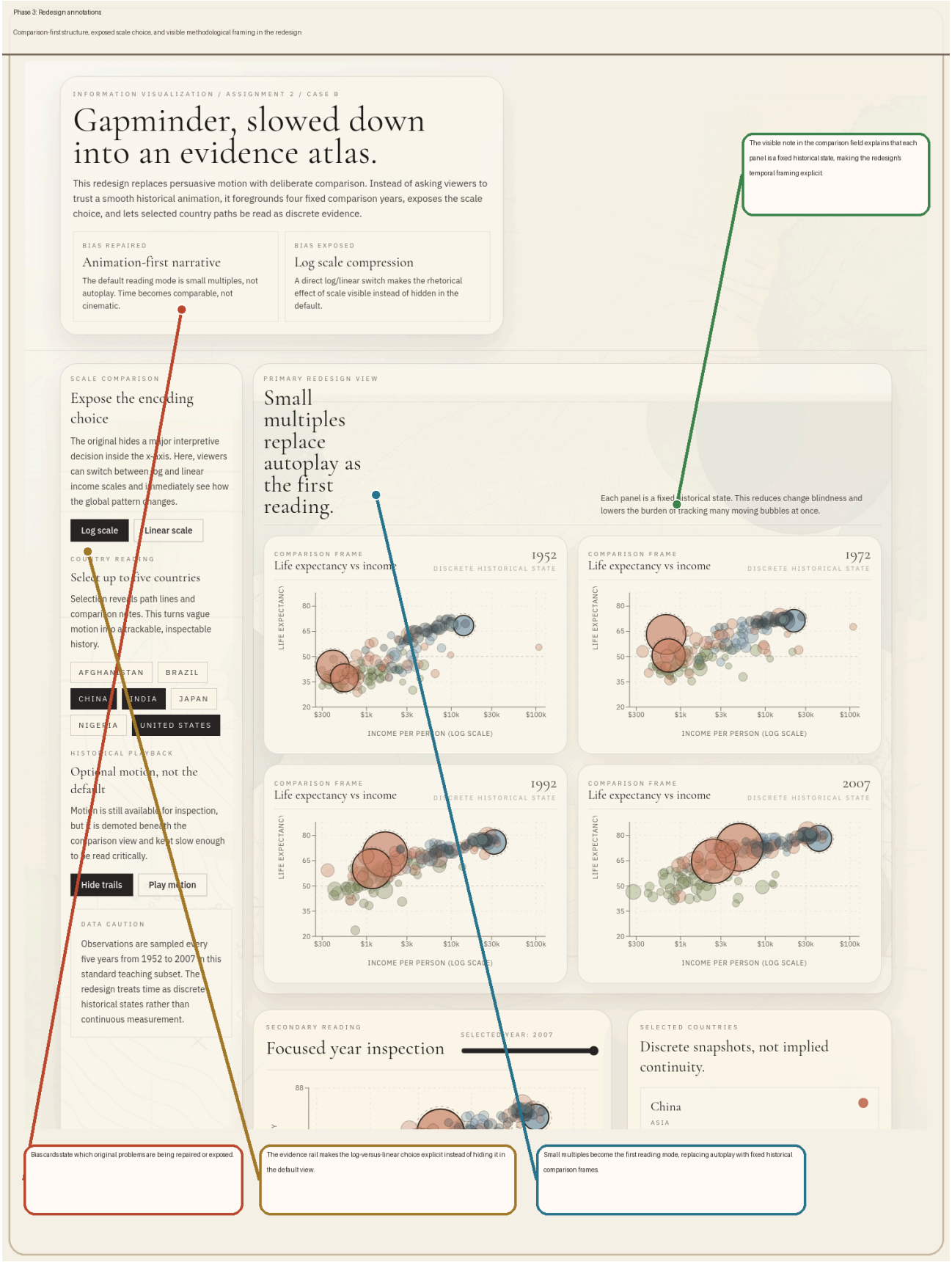
The second decision was to make the income axis a visible log/linear toggle. In the original interface, the logarithmic scale is analytically reasonable but rhetorically hidden. Making the scale switchable turns a concealed modeling assumption into something viewers can inspect.

The third decision was to include a visible method note explaining that the redesign remains faithful to Gapminder's core variables while using a five-year sampled teaching dataset. No synthetic observations were added, but the redesign makes the sampling structure explicit so the historical record does not appear smoother or more continuous than it actually is.

In Stage 3, I implemented and refined the interface as an evidence-focused reading environment. The page uses an archival atlas style with restrained typography, subtle drafting textures, and a left-side evidence rail. This was intentional, a glossy dashboard aesthetic would have recreated the same presentation energy the redesign aimed to reduce. During testing, I also compressed the hero section so the analytical content appears earlier above the fold. The first screen needed to signal comparison-first reading, rather than decorative atmosphere.

Figure 1

Annotated redesign screenshot showing explicit bias-repair framing, the exposed scale choice, the comparison-first structure, and the visible methodological note.



Before/After Comparison

The comparison below shows the same conceptual case before and after the redesign. In the original view, the interface encourages viewers to press play and watch a large field of moving bubbles, with a prominent year label creating a cinematic sense of progression. The logarithmic income axis is present, but its interpretive implications are not made explicit. In the redesign, the opening view presents four fixed comparison years instead of autoplay, allowing time to be examined side by side. The income axis is made transparent through a log/linear toggle, and a visible methodological note explains that the teaching dataset is sampled in five-year intervals. These changes address the key biases identified in Phase 1: animation-first storytelling, multiple-object tracking overload, hidden log-scale assumptions, and overstated continuity in the historical record.

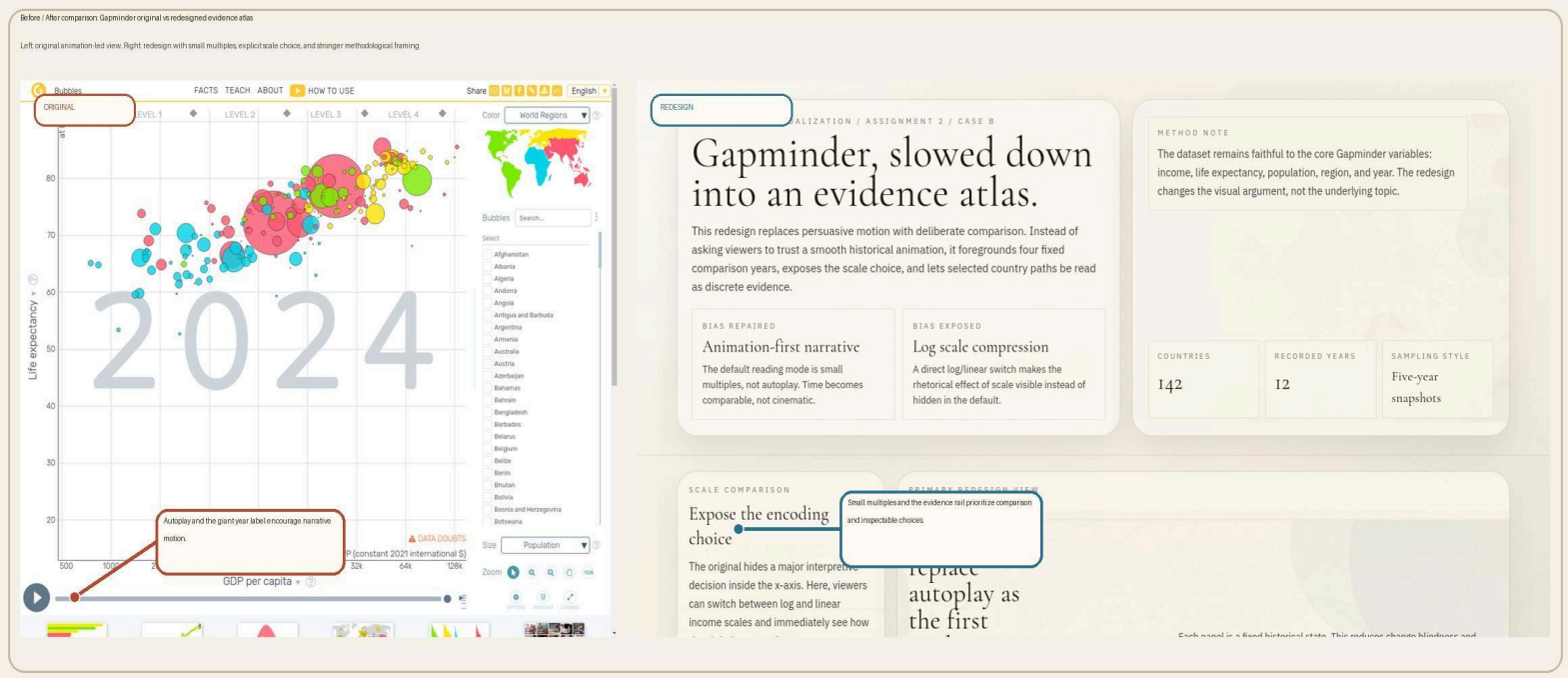


Figure 2 *Side-by-side comparison of the original Gapminder view and the redesigned evidence-atlas view, with annotations highlighting the most important changes.*

Self Audit

The redesign improves interpretability but does not remove all biases. First, bubble area still represents the population, so very large countries remain visually dominant. This is intentional, as the assignment requires preserving the original Gapminder variables, including population. While the interface shifts toward comparison and selective focus, the visual salience of large population bubbles remains.

Second, even with the log/linear toggle exposed, some viewers may still choose the log view without reflecting on how it changes interpretation. Making the assumption visible is better than hiding it, but visibility alone does not ensure critical reading.

Third, the redesign continues to use a scatterplot, so overlap and density can still be challenging when many countries cluster in similar positions.

If I had more time, the next improvement would be adding a stronger uncertainty layer for historical sampling and data quality. Currently, the redesign communicates five-year snapshot sampling through text and framing, but it does not visually indicate confidence, interpolation, or missing data for each country-year. Implementing this properly would require additional metadata and a design that conveys uncertainty without distracting from the main comparison task. I chose not to include it in the current assignment because a poorly justified uncertainty overlay could introduce confusion rather than improve interpretation.

One alternative I considered was a story-driven scrollytelling design that guides viewers through selected countries or regions with curated narrative checkpoints. This approach could provide a clearer, more authored explanation and stronger takeaways for novice audiences. However, it would also conflict with the main goal of the redesign. It would replace one persuasive narrative with another, rather than making the chart more inspectable and comparison-focused.

References

1. Assignment Brief

2. Lecture 10 Slides

3. Plotly Gapminder dataset.

https://github.com/plotly/datasets/blob/master/gapminder_with_codes.csv